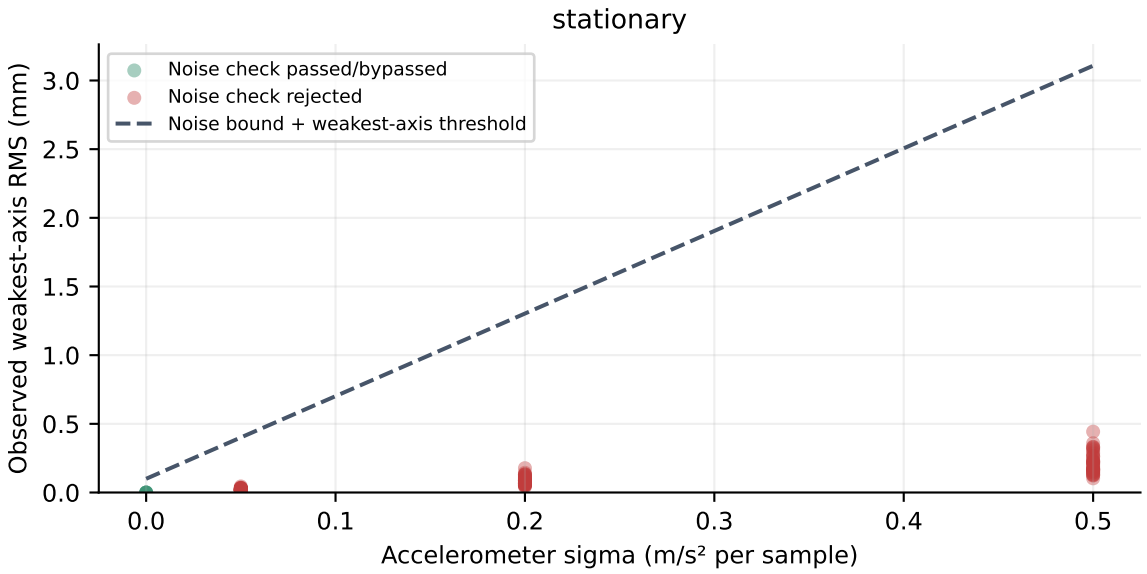
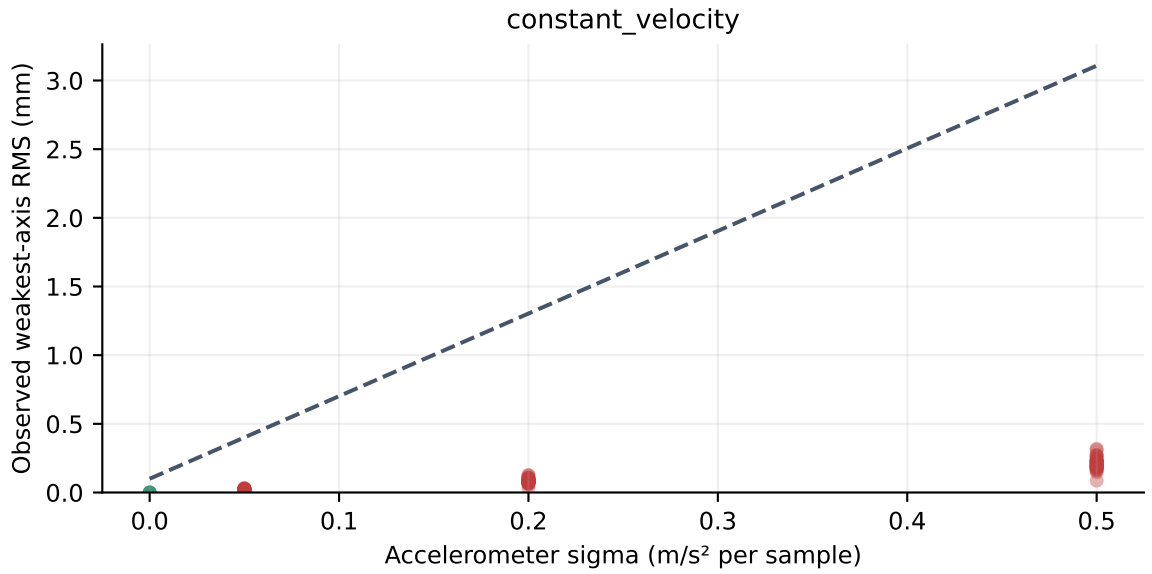
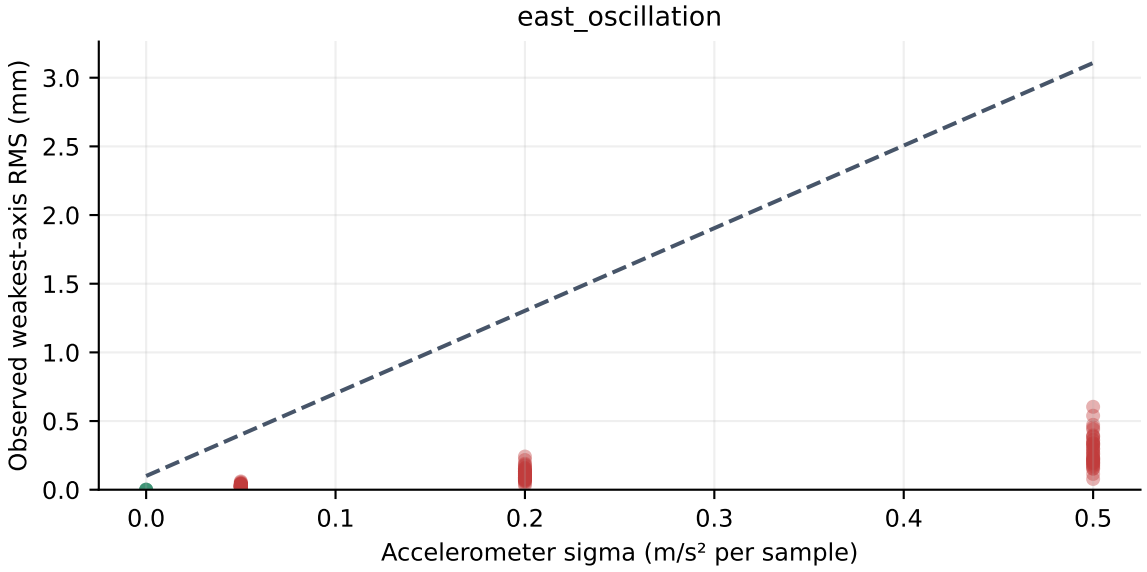
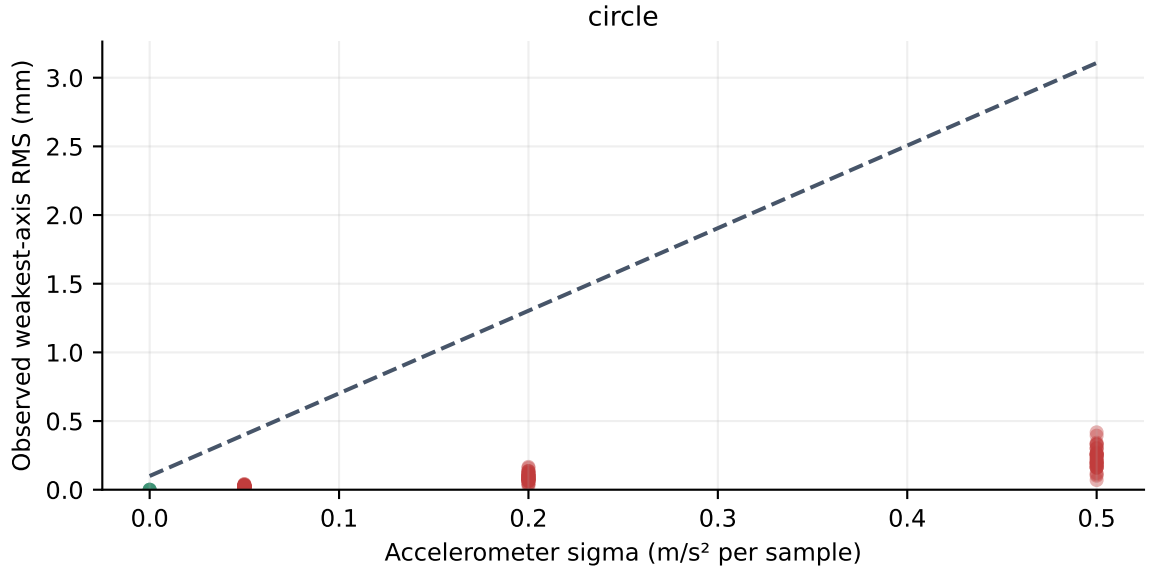
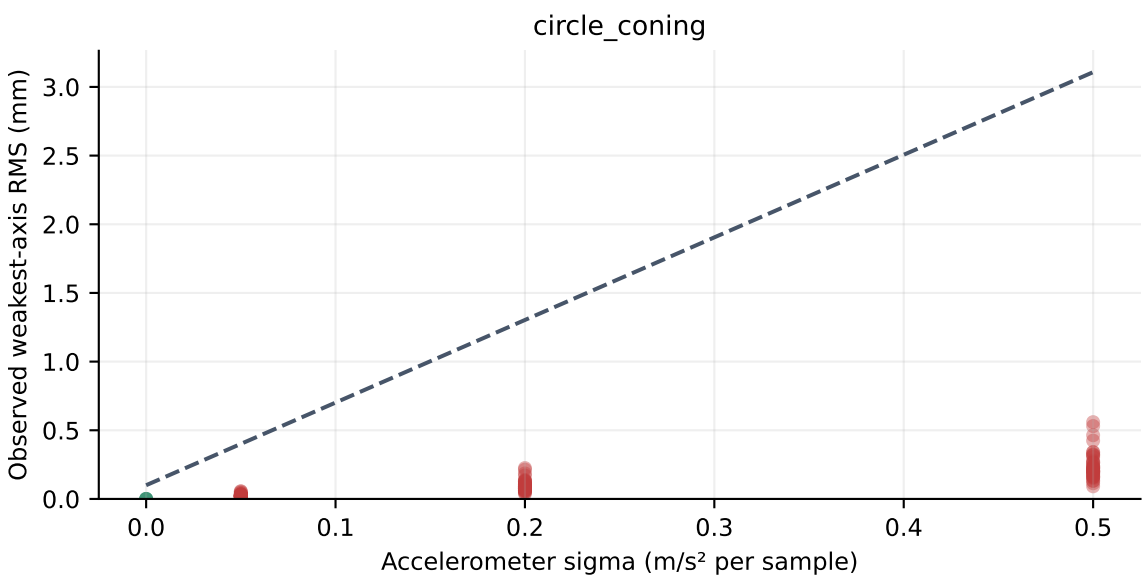
Observed excitation versus propagated white-noise uncertainty
Noise gate also checks total RMS and condition; green does not imply a conclusive detector decision

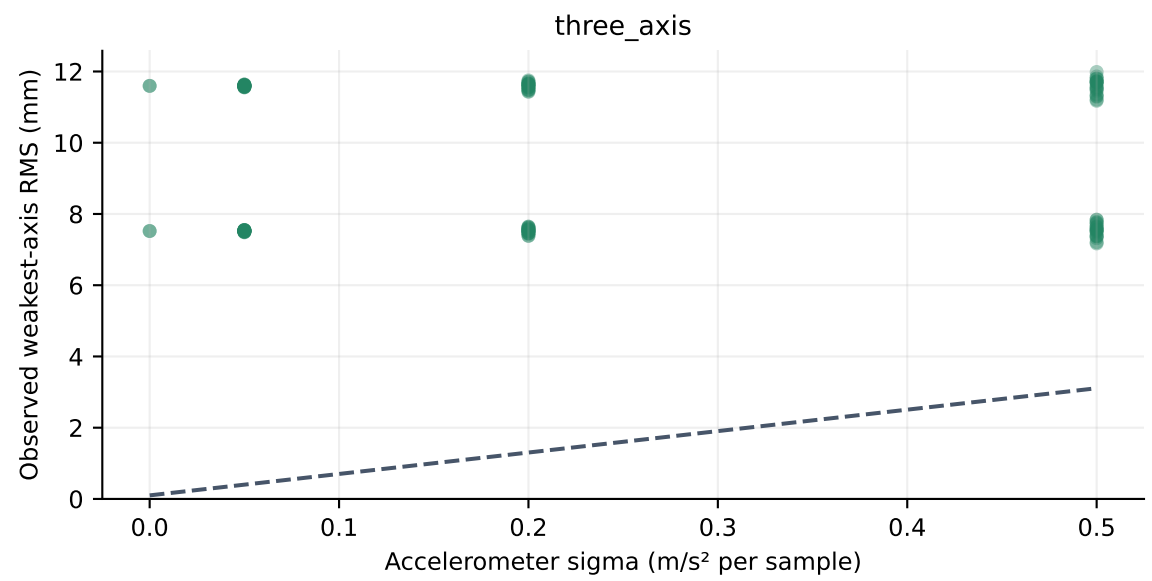


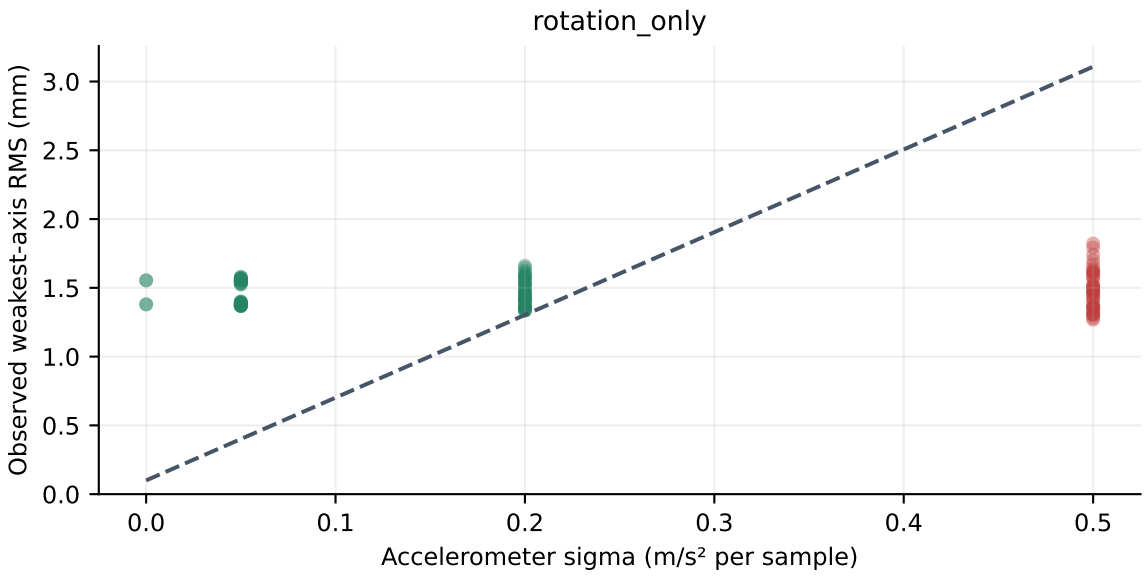


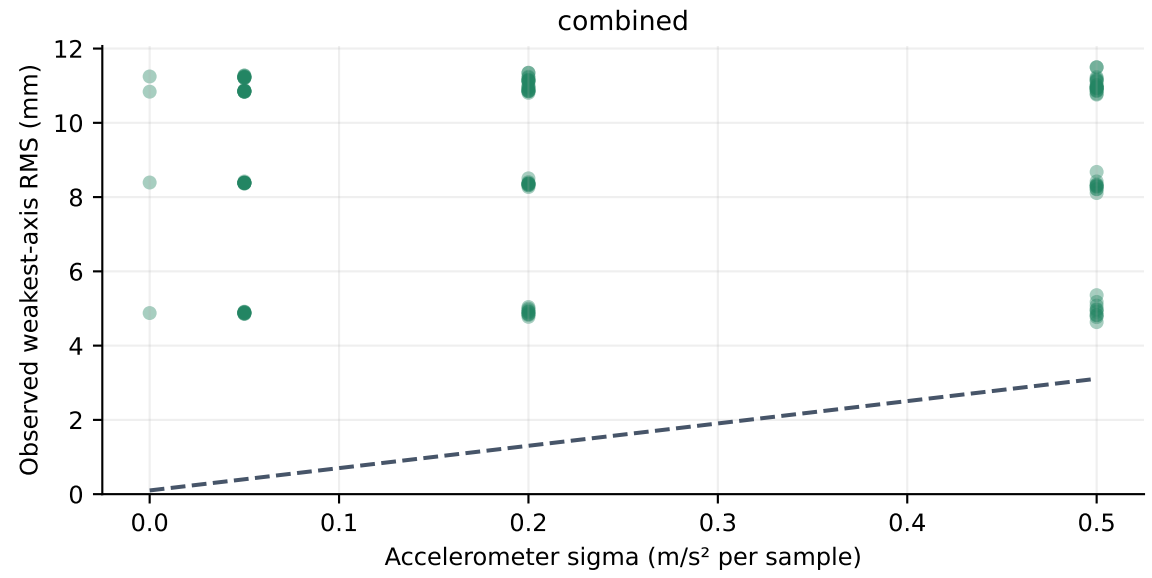


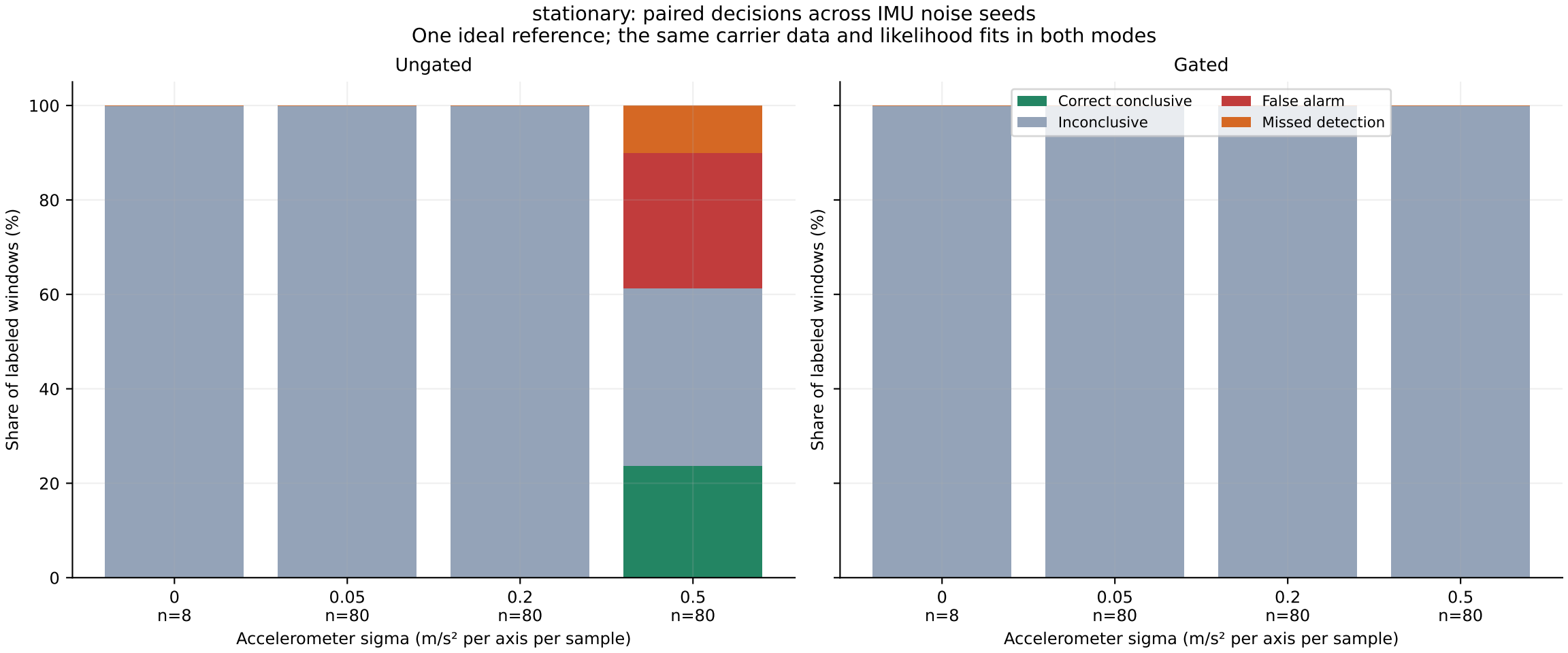


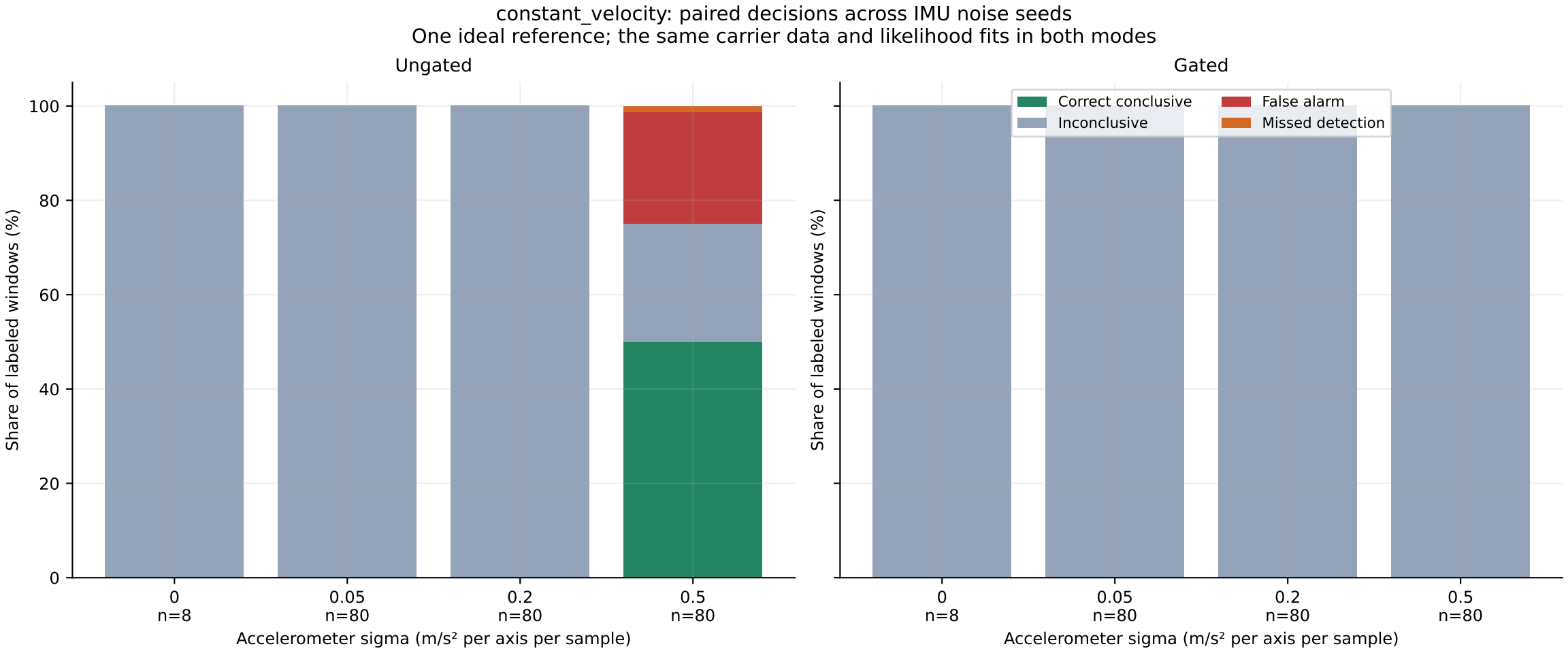


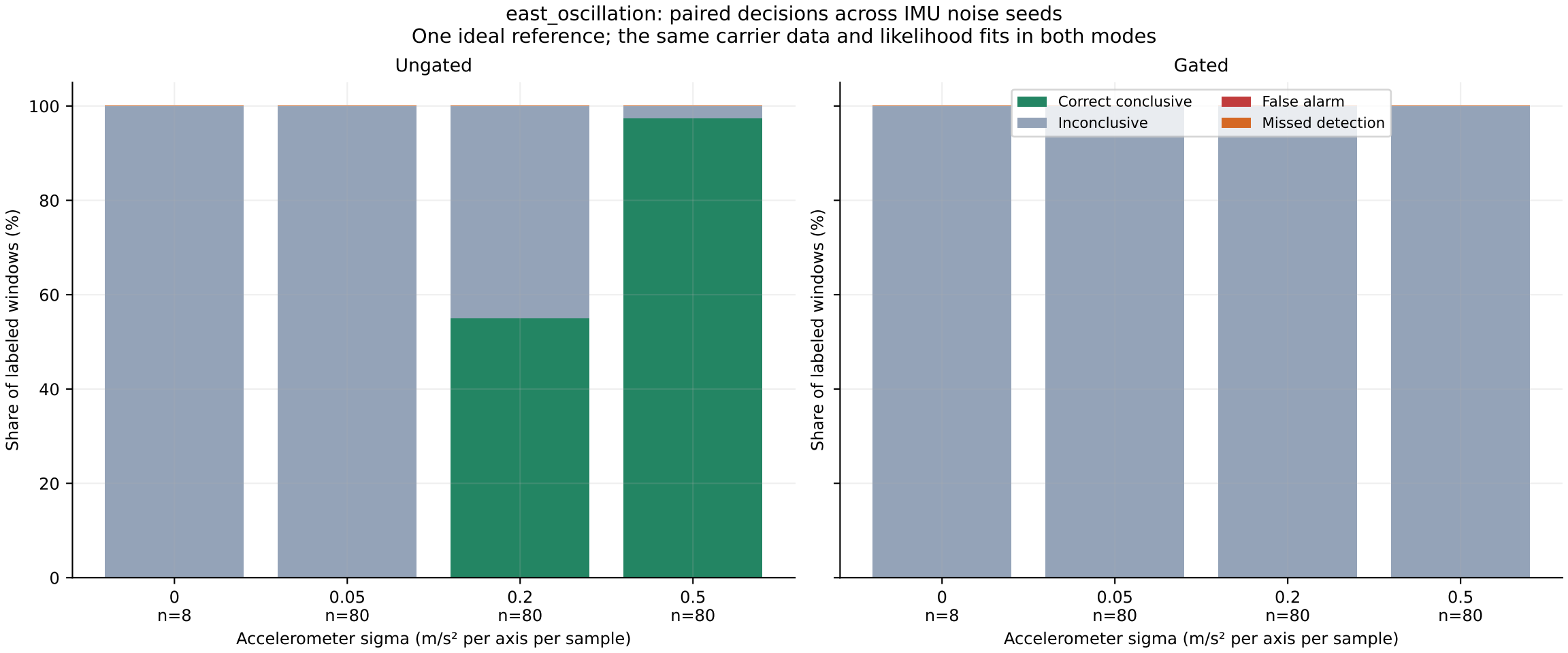


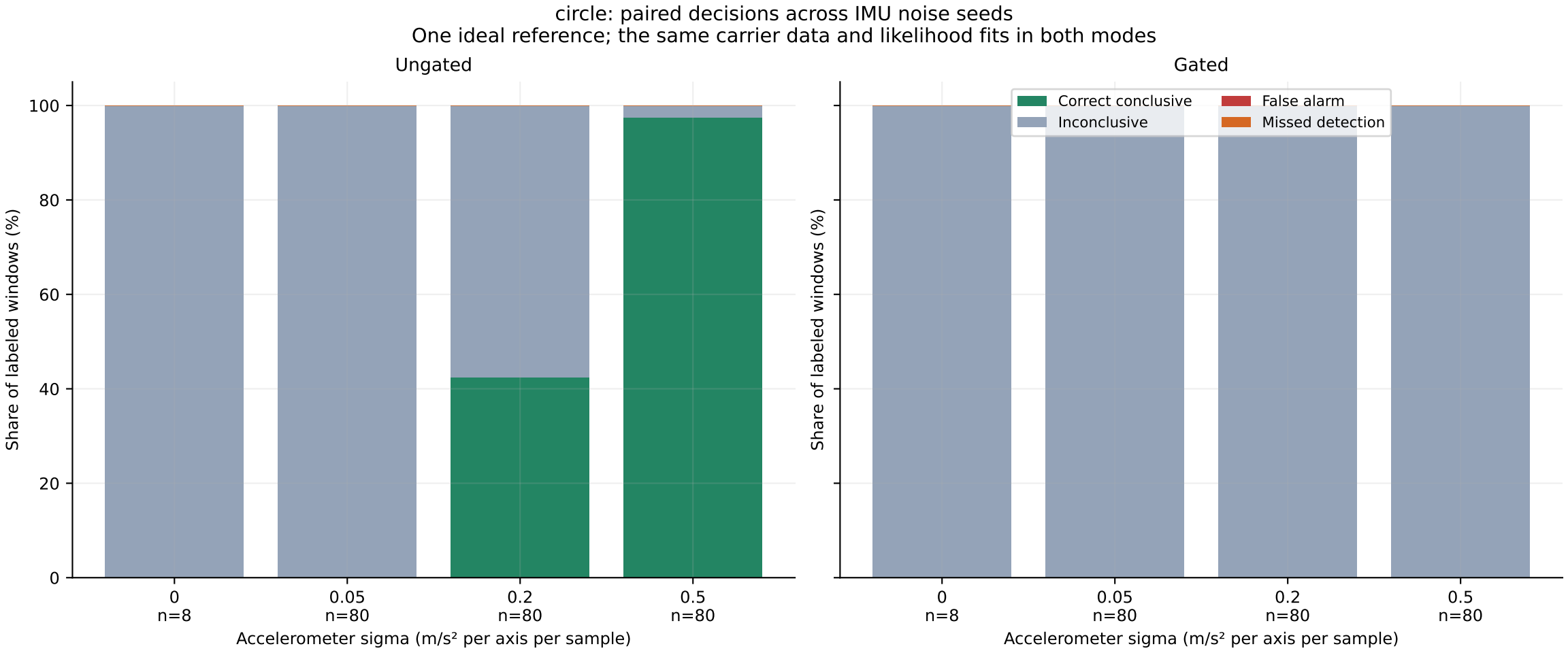


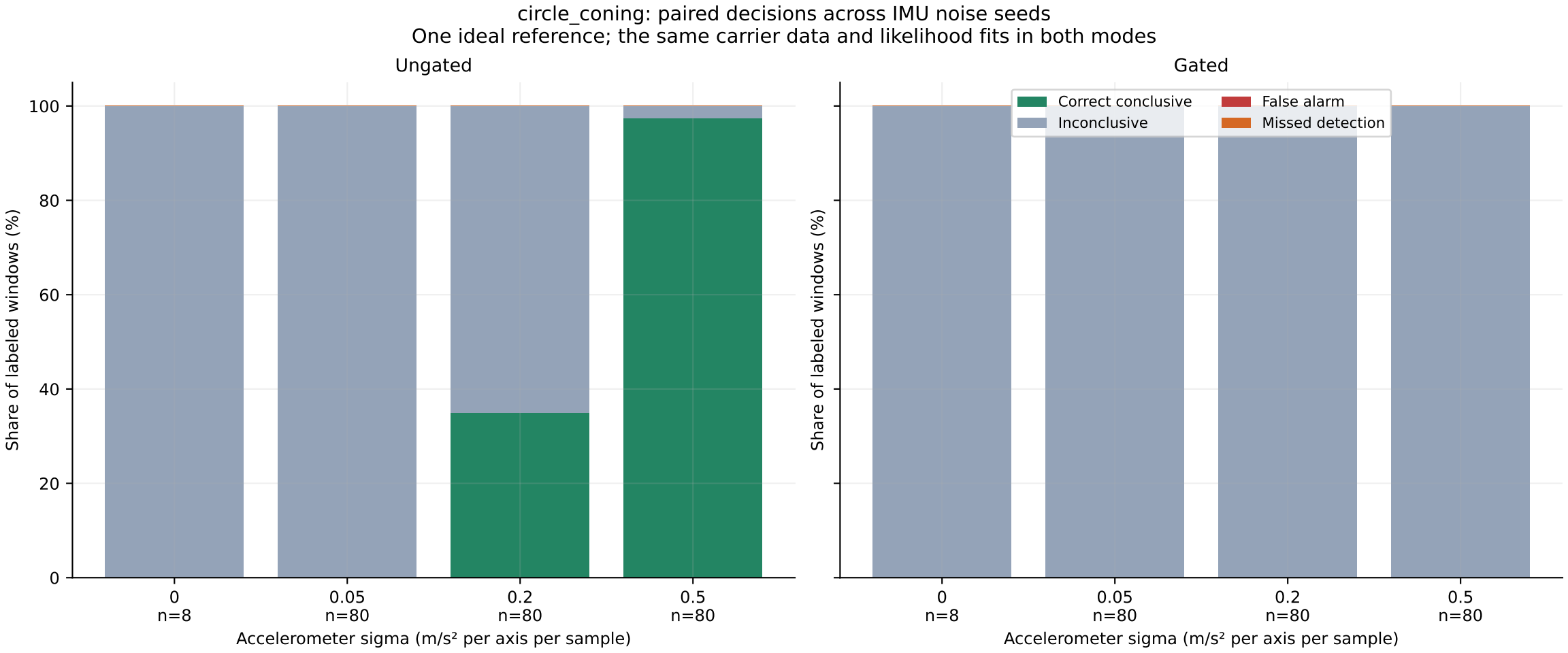


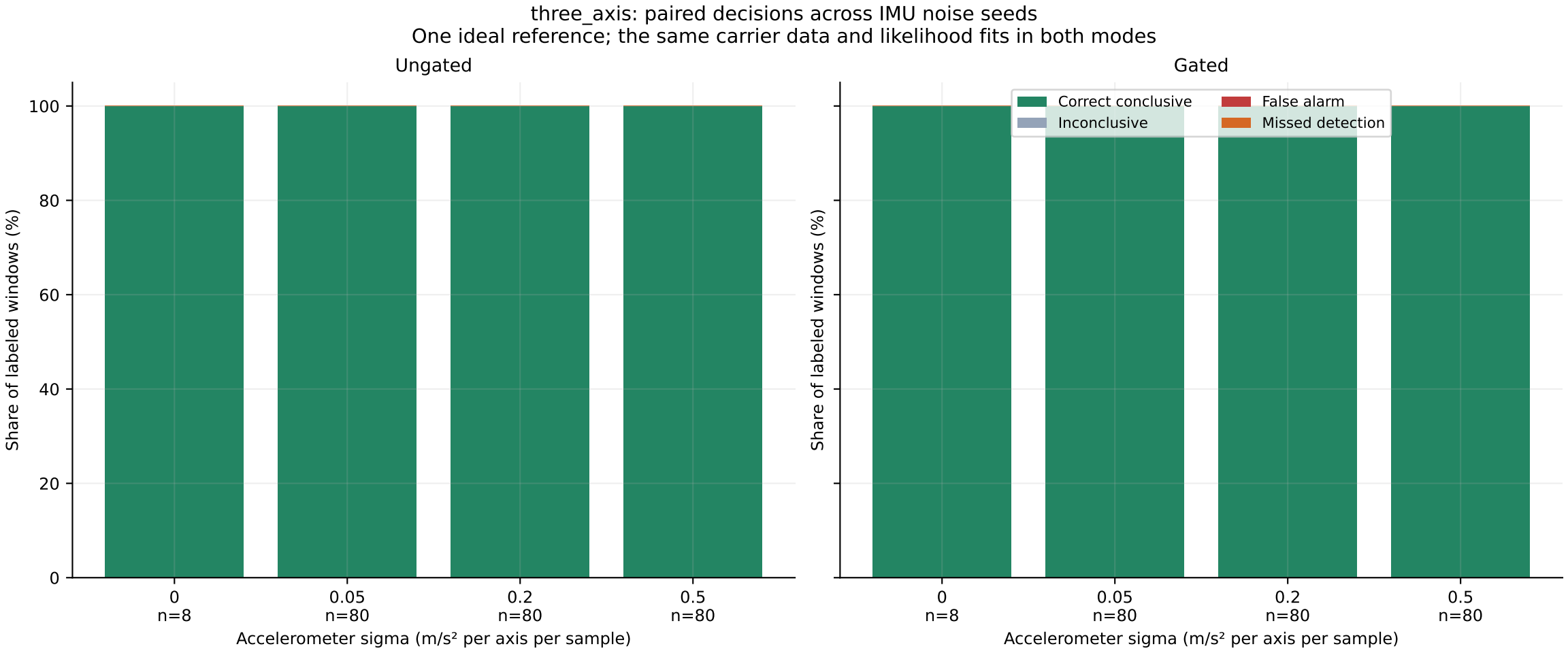

stationary: paired decisions across IMU noise seeds
One ideal reference; the same carrier data and likelihood fits in both modes
Ungated
Gated
Correct conclusive
Inconclusive
False alarm
Missed detection
Share of labeled windows (%)
0
20
40
60
80
100
0
n=8
0.05
n=80
0.2
n=80
0.5
n=80
Accelerometer sigma (m/s² per axis per sample)

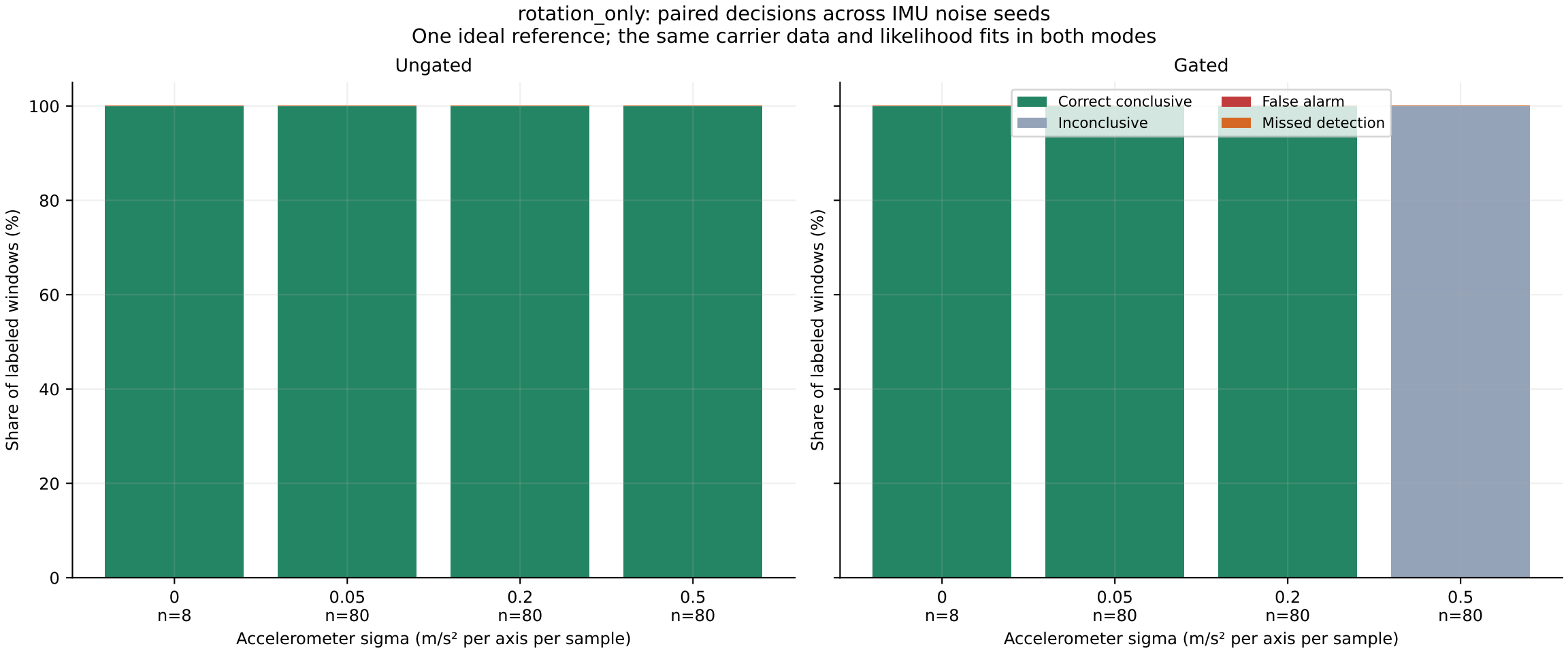

constant_velocity: paired decisions across IMU noise seeds
One ideal reference; the same carrier data and likelihood fits in both modes
Ungated
Gated
Correct conclusive
Inconclusive
False alarm
Missed detection
Share of labeled windows (%)
0
20
40
60
80
100
0
n=8
0.05
n=80
0.2
n=80
0.5
n=80
Accelerometer sigma (m/s² per axis per sample)

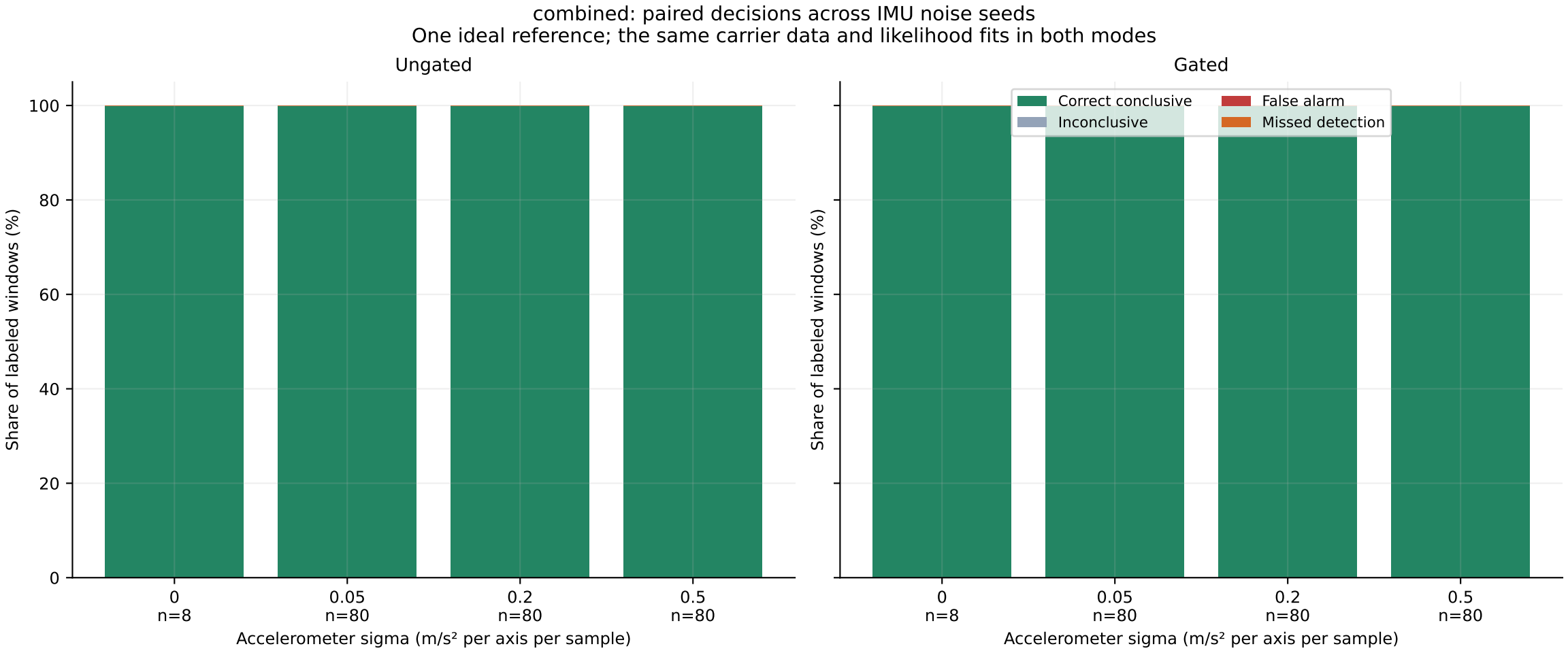

east_oscillation: paired decisions across IMU noise seeds
One ideal reference; the same carrier data and likelihood fits in both modes
Ungated
Gated
Correct conclusive
Inconclusive
False alarm
Missed detection
Share of labeled windows (%)
0
20
40
60
80
100
0
n=8
0.05
n=80
0.2
n=80
0.5
n=80
Accelerometer sigma (m/s² per axis per sample)


circle: paired decisions across IMU noise seeds
One ideal reference; the same carrier data and likelihood fits in both modes
Ungated
Gated
Correct conclusive
Inconclusive
False alarm
Missed detection
Share of labeled windows (%)
0
20
40
60
80
100
0
n=8
0.05
n=80
0.2
n=80
0.5
n=80
Accelerometer sigma (m/s² per axis per sample)


circle_coning: paired decisions across IMU noise seeds
One ideal reference; the same carrier data and likelihood fits in both modes
Ungated
Gated
Correct conclusive
Inconclusive
False alarm
Missed detection
Share of labeled windows (%)
0
20
40
60
80
100
0
n=8
0.05
n=80
0.2
n=80
0.5
n=80
Accelerometer sigma (m/s² per axis per sample)


three_axis: paired decisions across IMU noise seeds
One ideal reference; the same carrier data and likelihood fits in both modes
Ungated
Gated
Correct conclusive
Inconclusive
False alarm
Missed detection
Share of labeled windows (%)
0
20
40
60
80
100
0
n=8
0.05
n=80
0.2
n=80
0.5
n=80
Accelerometer sigma (m/s² per axis per sample)


rotation_only: paired decisions across IMU noise seeds
One ideal reference; the same carrier data and likelihood fits in both modes
Ungated
Gated
Correct conclusive
False alarm
Inconclusive
Missed detection
Share of labeled windows (%)
0
20
40
60
80
100
0
n=8
0.05
n=80
0.2
n=80
0.5
n=80
Accelerometer sigma (m/s² per axis per sample)


combined: paired decisions across IMU noise seeds
One ideal reference; the same carrier data and likelihood fits in both modes
Ungated
Gated
Correct conclusive
Inconclusive
False alarm
Missed detection
Share of labeled windows (%)
100
80
60
40
20
0
0
n=8
0.05
n=80
0.2
n=80
0.5
n=80
Accelerometer sigma (m/s² per axis per sample)